Math 401 Review Problems for Exam 1

Evaluate the following integrals

a) $\int_0^2 (x-2)^2 dx$

b) $\int \frac{2x^2 - 7x + 3}{3x^2} dx$

c) $\int \frac{4\sec^2 x}{\tan x} dx$

d) $\int \frac{\cos x e^{\sin x}}{\sqrt{e^{\sin x}}} dx$

e) $\int x\left(x^2 + 4\right)^2 dx$

f) $\int \frac{1 + x^5}{x^4} dx$

g) $\int \frac{x^4}{1 + x^5} dx$

h) $\int \frac{x^2}{\sqrt{1 + x^3}} dx$

i) $\int \frac{1}{(1 + x^2)\tan^{-1} x} dx$

2) Region R is bounded by $y = x^2$ and $y = 3x$. SET UP integrals that could be used to find the volume of the solid formed if R is revolved about:
a) the x –axis b) the y-axis c) $y = -4$ d) $x = 3$

3) Calculate the volume obtained by revolving the region bounded by $y = x^2$, $y = -2x + 8$, and $x = 0$ about the y – axis using the:
a) Disk/Washer Method b) Shell Method

4) SET UP integrals to evaluate the volume obtained by revolving the region bounded by $y = 4x - x^2$ and $y = 0$ about: a) the y –axis b) $x = 10$ c) $x = 2$

5) Use the Disk Method to verify the volume of a cone $= \frac{1}{3}\pi r^2 h$

6) The base of a solid is a circle with radius 4. Find the volume of the solid if cross-sections taken perpendicular to the base are:
a) squares b) equilateral triangles c) semi circles

7) A water tank shaped like a hemisphere with radius 12 ft is filled with water to a depth of 8 ft. Find the volume of water in the tank.

8 Hooke's Law states that the force required to maintain a spring stretched x units beyond its natural length is F = kx where k is a positive constant.
Suppose a force of 40 Newtons is required to hold a spring that has been stretched from its natural length of 1m to a length of 1.5m. How much work is done in stretching the spring from 1.5m to 2m?

9) A heavy rope, 50m long, weighs 0.5 lb/m hangs over the edge of a building 120 meters high. How much work is done in pulling the rope to the top of the building?

10) The vertical ends of a watering trough are semicircles of radius 3 meters. If the trough is 6 meters long, find the work done in emptying the trough by pumping the water to the top of the trough when the trough is full.

11) The vertical ends of a storage trough are isosceles trapezoids with dimensions of 6 meter at the top, and 4 meters at the bottom. The trough is 8 meters high and 10 meters long. Find the work done in emptying the trough if the trough contains olive oil and it is filled: The density of olive oil is 890 kg/m^3

a) filled to the top. b) filled to within 2 meters from the top.